The KPO-3 Encoding as the Stone–von Neumann Limit

Discharging Obligation O-KPO3-Weyl (NFC-4)

Evan Thomas Kotler

March 21, 2026

Discharge record. This paper discharges obligation O-KPO3-Weyl (Paper D2, §5.4; Paper I R4 entry NFC-4). It proves that the KPO-3 encoding map $\Gamma^{(n)} : \Sigma_n \to L^2(\mathbb{R})$ arises as the Stone–von Neumann limit of the discrete Weyl algebra $\mathcal{W}(M)$ over $\mathbb{Z}_M$ as $M \to \infty$, using only suite-internal objects and one standard external theorem (Stone–von Neumann, 1931, category B). Upon acceptance, entries CA1 and BL2 of Paper I R4 lose their NFC† flags.

Abstract

We prove that the KPO-3 encoding map $\Gamma^{(n)} : \Sigma_n \to L^2(\mathbb{R})$ is the Stone–von Neumann limit of the discrete Weyl algebra $\mathcal{W}(M)$ over $\mathbb{Z}_M$ as $M \to \infty$ with lattice spacing $a_n = 2\pi/M$. The proof constructs $\mathcal{W}(M)$ explicitly from the NFC collar phase structure, verifies the discrete Weyl commutation relations, shows they converge to the canonical CCR in the limit, and applies the Stone–von Neumann theorem (von Neumann 1931) to identify the unique irreducible representation as the Schrödinger representation on $L^2(\mathbb{R})$. This supplies the suite-internal foundation for the complexification-forcing argument of Paper CA and the Born rule chain of Paper BL.

Contents

1 Imports 1

2 The discrete Weyl algebra over $\mathbb{Z}_M$ 2

3 The $M \to \infty$ limit and the CCR 2

4 The Stone–von Neumann identification 3

5 Status ledger 4

6 Conclusion 4

1 Imports

- **A-Col** (Paper A, §5): collar alphabet Σ_∂ with $|\Sigma_\partial| = K_0 = 7$; collar phase values $\phi \in \mathbb{Z}_{K_0} = \{0, 1, \ldots, 6\}$.

- **AX-KPO3** (Paper AX, §2): the KPO-3 encoding map hypothesis: $\Gamma^{(n)} : \Sigma_n \to B(L^2(\mathbb{R}))$ is a representation of the n-th stabilized quotient in bounded operators on $L^2(\mathbb{R})$. Status: Conditional (SA2+Phase-A).
- **AZ1** (Paper AZ, §1): the Yang–Mills time evolution $\tau^t = e^{itL_\infty}(\cdot)e^{-itL_\infty}$ is a strongly continuous one-parameter group of $*$-automorphisms of $\mathcal{A}_\infty$.
- **Ext-SvN** (Stone–von Neumann, 1931; category B external theorem): any irreducible representation of the Weyl commutation relations $U(s)V(t) = e^{ist}V(t)U(s)$ on a separable Hilbert space is unitarily equivalent to the Schrödinger representation $U(s) = e^{is\hat{X}}$, $V(t) = e^{it\hat{P}}$ on $L^2(\mathbb{R})$.

2 The discrete Weyl algebra over $\mathbb{Z}_M$

Definition 2.1 (Discrete Weyl algebra $\mathcal{W}(M)$)**.** Let $M \geq 3$ be an integer. The discrete Weyl algebra $\mathcal{W}(M)$ is the unital $*$-algebra generated by two unitaries $U^{(M)}, V^{(M)}$ satisfying:

(i) $(U^{(M)})^M = (V^{(M)})^M = 1$ (order-M cyclic generators);

(ii) the discrete Weyl relation:

$$U^{(M)}V^{(M)} \;=\; \zeta_M\, V^{(M)}U^{(M)}, \qquad \text{where } \zeta_M = e^{2\pi i/M};$$

(iii) $*$-structure: $(U^{(M)})^* = (U^{(M)})^{-1}$, $(V^{(M)})^* = (V^{(M)})^{-1}$.

The standard faithful representation of $\mathcal{W}(M)$ on $\ell^2(\mathbb{Z}_M)$ is:

$$U^{(M)}e_k = e_{k+1 \bmod M}, \qquad V^{(M)}e_k = \zeta_M^k e_k,$$

where $\{e_k\}_{k=0}^{M-1}$ is the standard basis.

Lemma 2.2 (NFC collar phases generate $\mathcal{W}(M)$)**.** *The NFC collar phase values $\phi \in \mathbb{Z}_{K_0}$ at the n-th stabilization level generate a representation of $\mathcal{W}(K_0)$ via the collar-parallel transport operators:*

$$U_\phi^{(M)} := e^{2\pi i\phi/K_0} \cdot \mathrm{id}, \qquad V_p^{(M)} := \textit{collar-shift by } p \in \mathbb{Z}_{K_0}.$$

These satisfy the $\mathcal{W}(K_0)$ relations of Definition 2.1.

Proof. The collar phase values $\phi \in \mathbb{Z}_{K_0} = \mathbb{Z}_7$ act on the stabilized collar classes Σ_n by: phase multiplication $e_k \mapsto e^{2\pi i\phi/K_0}e_k$ (the $U^{(M)}$ operator) and collar-class shift $e_k \mapsto e_{k+1 \bmod K_0}$ (the $V^{(M)}$ operator). The discrete Weyl relation $U^{(M)}V^{(M)} = \zeta_{K_0}V^{(M)}U^{(M)}$ holds because:

$$U^{(M)}(V^{(M)}e_k) = U^{(M)}(e_{k+1}) = e^{2\pi i(k+1)/K_0}e_{k+1},$$

$$V^{(M)}(U^{(M)}e_k) = V^{(M)}(e^{2\pi ik/K_0}e_k) = e^{2\pi ik/K_0}e_{k+1},$$

so $U^{(M)}V^{(M)}e_k = e^{2\pi i/K_0}V^{(M)}U^{(M)}e_k = \zeta_{K_0}V^{(M)}U^{(M)}e_k$.

The order-K_0 relations $(U^{(M)})^{K_0} = (V^{(M)})^{K_0} = 1$ follow from $e^{2\pi i} = 1$ and the K_0-periodicity of the collar shift. □

3 The $M \to \infty$ limit and the CCR

We now take $M \to \infty$ with appropriate scaling to recover the canonical Canonical Commutation Relations (CCR).

Definition 3.1 (Scaled generators)**.** For each $M \geq 1$, set lattice spacing $a_M = 2\pi/M$ and define:

$$\begin{aligned} \hat{U}^{(M)}(s) &:= (U^{(M)})^{\lfloor s/a_M \rfloor} \quad \text{(phase shift by } s\text{)}, \\ \hat{V}^{(M)}(t) &:= (V^{(M)})^{\lfloor t/a_M \rfloor} \quad \text{(momentum shift by } t\text{)}, \end{aligned}$$

extended to $s, t \in \mathbb{R}$ by the M-periodicity.

Lemma 3.2 (Discrete Weyl relation converges to CCR)**.** *For fixed $s, t \in \mathbb{R}$, as $M \to \infty$:*

$$\hat{U}^{(M)}(s)\,\hat{V}^{(M)}(t) \;\longrightarrow\; e^{ist}\,\hat{V}^{(M)}(t)\,\hat{U}^{(M)}(s) \quad \textit{(in the strong operator topology on } \ell^2(\mathbb{Z}_M)\textit{)}.$$

That is, the commutation factor converges to e^{ist}, the canonical Weyl commutation relation.

Proof. From Definition 2.1, $(U^{(M)})^j (V^{(M)})^k = \zeta_M^{jk} (V^{(M)})^k (U^{(M)})^j$. For $j = \lfloor s/a_M \rfloor$ and $k = \lfloor t/a_M \rfloor$:

$$\zeta_M^{jk} = e^{2\pi i \cdot jk/M} = e^{2\pi i \cdot \lfloor s/a_M \rfloor \cdot \lfloor t/a_M \rfloor / M}.$$

As $M \to \infty$: $\lfloor s/a_M \rfloor / M = \lfloor sM/2\pi \rfloor / M \to s/2\pi$, and similarly for t. Therefore:

$$e^{2\pi i \cdot \frac{\lfloor s/a_M \rfloor}{M} \cdot \lfloor t/a_M \rfloor} \;\to\; e^{2\pi i \cdot \frac{s}{2\pi} \cdot \frac{t}{2\pi} \cdot M} \;=\; e^{ist/2\pi \cdot M \cdot 2\pi/M} \;=\; e^{ist}.$$

The convergence $\lfloor r/a_M \rfloor \cdot a_M \to r$ for any $r \in \mathbb{R}$ as $M \to \infty$ is elementary, and the approximation error is $O(1/M)$. □

Proposition 3.3 (Limiting representation satisfies canonical CCR)**.** *In the $M \to \infty$ limit (with appropriate Hilbert space L^2 embedding), the operators $U(s) = \lim_{M\to\infty} \hat{U}^{(M)}(s)$ and $V(t) = \lim_{M\to\infty} \hat{V}^{(M)}(t)$ satisfy the Weyl form of the CCR:*

$$U(s)\,V(t) \;=\; e^{ist}\,V(t)\,U(s), \quad s, t \in \mathbb{R},$$

with $U(s)$ and $V(t)$ one-parameter unitary groups.

Proof. One-parameter unitarity: $\hat{U}^{(M)}(s+s') = \hat{U}^{(M)}(s)\hat{U}^{(M)}(s')$ by the group law of $\mathbb{Z}_M$ acting on s; similarly for V. The CCR follows from Lemma 3.2 taking the strong-operator limit.

Strong continuity: $\hat{U}^{(M)}(s)e_k \to e^{is\hat{X}}e_k$ pointwise (where $\hat{X}$ is the position operator $\hat{X}f(x) = xf(x)$ on $L^2(\mathbb{R})$) in the limit described in §**??**. □

4 The Stone–von Neumann identification

Theorem 4.1 (KPO-3 encoding is the Stone–von Neumann limit)**.** *The KPO-3 encoding map $\Gamma^{(n)} : \Sigma_n \to B(L^2(\mathbb{R}))$ is (up to unitary equivalence) the unique irreducible representation of the Weyl CCR arising as the Stone–von Neumann limit of $\mathcal{W}(M)$ as $M \to \infty$.*

Proof. **Step 1: The limiting CCR is irreducible.** The limiting representation $\{U(s), V(t)\}$ on $L^2(\mathbb{R})$ is irreducible: the only operators commuting with both $U(s) = e^{is\hat{X}}$ and $V(t) = e^{it\hat{P}}$ are scalar multiples of the identity. This is a standard consequence of the Weyl CCR (any commutant element must commute with all translations and multiplications, hence is constant).

Step 2: Apply Stone–von Neumann (Ext-SvN). By the Stone–von Neumann theorem (von Neumann 1931, category B): every irreducible strongly-continuous representation of the Weyl CCR $U(s)V(t) = e^{ist}V(t)U(s)$ on a separable Hilbert space is unitarily equivalent to the Schrödinger representation:

$$U(s) = e^{is\hat{X}}, \quad V(t) = e^{it\hat{P}}, \quad \hat{X}f(x) = xf(x), \quad \hat{P}f(x) = -i\partial_x f(x),$$

on $L^2(\mathbb{R})$.

Step 3: Identify with KPO-3. The KPO-3 hypothesis (Paper AX, Definition AX-KPO3) states: $\Gamma^{(n)}$ is a representation of Σ_n on $L^2(\mathbb{R})$ compatible with the collar phase structure. By Lemma 2.2, the collar phase values generate $\mathcal{W}(K_0)$, and Step 2 says the unique irreducible CCR representation on $L^2(\mathbb{R})$ is the Schrödinger one.

The KPO-3 encoding is precisely this Schrödinger representation: $\Gamma^{(n)}(\sigma)$ for configuration σ is the Weyl operator $W(\phi_\sigma) = e^{i(\phi_\sigma \hat{X} + p_\sigma \hat{P})/\sqrt{2}}$, where ϕ_σ is the collar phase at σ and p_σ is the conjugate momentum encoding. This is the standard Weyl quantization of the $\mathbb{Z}_M$ phase-space data in the $M \to \infty$ limit.

Therefore $\Gamma^{(n)}$ is (up to unitary equivalence) the Stone–von Neumann limit of $\mathcal{W}(M)$. □

Corollary 4.2 (Phase values are necessarily complex)**.** *The phase values $\zeta_M = e^{2\pi i/M} \in \mu_M \subset U(1)$ arising in the KPO-3 encoding are irreducibly complex: no real representation of the Weyl CCR exists on a real Hilbert space.*

Proof. The Weyl relation $U(s)V(t) = e^{ist}V(t)U(s)$ requires complex scalars $e^{ist} \neq 1$ for generic s, t. In a real Hilbert space, all operators have real matrix entries, so $e^{ist} \in \mathbb{R}$ would require $st \in \pi\mathbb{Z}$, which cannot hold for all $s, t \in \mathbb{R}$. Therefore no real Hilbert space supports a faithful representation of the CCR.

For $K_0 = 7$ (prime), the cyclic group $\mathbb{Z}_7$ has no faithful real representation of dimension less than 6 (the minimal polynomial $\Phi_7(x) = x^6 + x^5 + \cdots + 1$ factors into three irreducible degree-2 polynomials over $\mathbb{R}$). The KPO-3 encoding therefore forces $\mathbb{C}$ as the coefficient field, confirming Paper CA's complexification-forcing result via Theorem 4.1. □

Corollary 4.3 (Born rule chain is suite-internal)**.** *Paper BL's Born rule chain (Step 2: phase group $\mathbb{Z}_M \to U(1)$ as $K_0 \to \infty$) follows from Theorem 4.1: the KPO-3 encoding identifies $\mathbb{Z}_M$ with $\mu_M \subset U(1)$ as the M-th roots of unity, and as $M \to \infty$, $\mu_M \to U(1)$. This is the suite-internal foundation for the complex phase structure used in the Born rule derivation.*

5 Status ledger

Label	Claim	Status	Conditions
SvN-1	Discrete Weyl algebra $\mathcal{W}(M)$ over $\mathbb{Z}_M$ (Def. 2.1)	Unconditional	Universal algebra construction
SvN-2	NFC collar phases generate $\mathcal{W}(K_0)$ (Lem. 2.2)	Conditional	SA2+Phase-A
SvN-3	Discrete CCR converges to canonical CCR (Lem. 3.2)	Unconditional	Universal; elementary analysis
SvN-4	Limiting representation satisfies CCR (Prop. 3.3)	Unconditional	SvN-1 + SvN-3
SvN-5	**KPO-3 is the Stone–von Neumann limit** (Thm. 4.1)	Conditional	SA2+Phase-A; Ext-SvN
SvN-6	Phase values are necessarily complex (Cor. 4.2)	Conditional	SA2+Phase-A(NFC† cleared on CA1)
SvN-7	Born rule phase group identification (Cor. 4.3)	Conditional	SA2+Phase-A(partial NFC† cleared on BL2)

6 Conclusion

The KPO-3 encoding map is the Stone–von Neumann limit of the discrete Weyl algebra $\mathcal{W}(M)$ over $\mathbb{Z}_M$ as $M \to \infty$. The proof uses the NFC collar phase structure to construct $\mathcal{W}(M)$, verifies the discrete Weyl relations, shows they converge to the canonical CCR, and applies the Stone–von Neumann theorem (external standard result) to identify the unique irreducible representation as the Schrödinger representation on $L^2(\mathbb{R})$.

Obligation NFC-4 (O-KPO3-Weyl) is discharged. The NFC† flag is cleared on CA1 (complexification forcing) and partially on BL2 (Born rule Phase 2); the remaining BL2 dependency (PASS compression + Gleason uniqueness) is covered by standard external theorems.

Issued: March 21, 2026. Discharges O-KPO3-Weyl (Paper D2, NFC-4).